Predicting Online Shopper Purchase Intention:

A Study of Feature Selection and Class-Imbalance Handling on Model Performance and Interpretability

FML Mini-Project Synopsis

Manipal Institute of Technology

Department of Data Science and Engineering

Semester 5

Submitted by:

Reg. No.	Name	Branch	Section
240968005	Nayantara Nambiar	DSE	B
240968100	Tarunashri Surapaneni	DSE	B
240968260	Dishita Sanvi Thogata	DSE	B

1. Abstract

E-commerce platforms convert only a small fraction of visiting sessions into purchases, making the early identification of high-intent visitors valuable for real-time interventions such as personalised offers or exit-intent prompts. This project uses the UCI Online Shoppers Purchasing Intention Dataset, comprising 12,330 browsing sessions described by 18 behavioural and administrative features, to build and compare machine learning models that classify a session as ending in a purchase (Revenue = True) or not. The dataset is strongly imbalanced, with only 15.5% of sessions resulting in a purchase, which is treated as a central modelling challenge rather than an inconvenience. Beyond training a single classifier, the project systematically studies how different feature-selection strategies and class-imbalance handling techniques jointly affect both predictive performance and model interpretability, an interaction that recent literature has not examined in a unified setting. Four models – Logistic Regression, Decision Tree, Random Forest, and Support Vector Machine – are trained under multiple feature-selection and resampling configurations, and evaluated using precision, recall, F1-score, and ROC-AUC rather than raw accuracy. The outcome is expected to be a reproducible pipeline and empirical evidence on which preprocessing choices best balance detection of the minority (purchasing) class against the explainability e-commerce teams need to act on model outputs.

2. Introduction and Motivation

Online retail generates continuous streams of behavioural data – page views, session duration, bounce and exit rates – that reflect a visitor's evolving intent well before a purchase decision is made. If this intent can be estimated in real time, businesses can intervene: showing a discount to a hesitant visitor, prioritising customer support for a high-value session, or adjusting page layout for browsers unlikely to convert. This makes purchase-intention prediction a practically important classification problem within e-commerce analytics.

However, most published work on this problem, including several 2023–2025 studies surveyed below, reports accuracy or F1 improvements from applying a single resampling technique or a single feature-selection method in isolation, without examining how these two preprocessing choices interact, or how that interaction affects the interpretability of the resulting model. Since real deployment requires both reliable minority-class detection and explanations that a marketing or product team can trust, this gap is directly relevant. This project is motivated by closing that gap on a well-understood, publicly available dataset, so that the experimental design – rather than data cleaning – can be the primary contribution.

3. Literature Survey

Ten recent and relevant studies (2023–2026) were reviewed, covering classical machine learning, ensemble methods, deep/graph-based approaches, resampling techniques, and interpretability-focused methods applied to purchase-intention or closely related e-commerce prediction problems. The comparison below summarises each study's method, dataset, evaluation metrics, and the specific gap it leaves open.

Study	Focus / Venue	Method(s)	Dataset	Metrics	Gap Identified
Wang et al. (2023)	XGBoost-based purchase prediction (Electronics)	XGBoost + feature importance ranking	Clickstream data (e-commerce)	Accuracy, F1	No systematic imbalance handling
Feature Eng. & Classification Study (2023)	Modeling purchase intention via feature transformation (ResearchGate)	Feature transformation + outlier detection + balancing	UCI Online Shoppers	Accuracy, Precision, Recall	Feature engineering tested ad hoc, not isolated from model choice
Zhou & Hudin (2024)	Time-series attention + GNN user profiling (PLOS ONE)	Attention + Graph Neural Network	E-commerce event/timestamp logs	AUC, F1	High model complexity; low interpretability
Arora, Sakshi & Gupta (2024)	Ensemble learning on oversampled data (Springer)	PART, DT, Naive Bayes, LR + SMOTE	UCI Online Shoppers	Accuracy, Precision, Recall, F1	Only oversampling explored; no feature-selection comparison
Purchase Intention via ML (2024)	Classical ML comparison study (conference/ResearchGate)	Logistic Regression, Decision Tree, Ensembles	UCI Online Shoppers	Accuracy	Imbalance not explicitly addressed
Alamuri & Bondalapu (2025)	SMOTE + Chi-square feature selection pipeline (ASEAN J. Sci. Tech. Reports)	SVC, Random Forest, Ridge Classifier + SMOTE + SelectKBest	UCI Online Shoppers	Accuracy, Precision, Recall, F1, ROC-AUC	Feature selection and imbalance handling combined but not decoupled for interpretability analysis
Tokuç & Dag (2025)	Clickstream representation comparison (IEEE Access)	Multiple ML models across data representations	Clickstream data	Accuracy, F1, AUC	Focuses on data representation, not imbalance-interpretability trade-off
Lin (2025)	Consumer behavior prediction for precision marketing (PLOS One)	SMOTE + variance filtering + ML classifiers	UCI Online Shoppers	Accuracy, F1	Feature filtering is variance-based only; no comparative feature-selection study
Deep Dive SLR (2025)	Systematic literature review, 47 studies 2019–2025 (Springer, Machine Learning)	Review of feature engineering & imbalance methods	Multiple e-commerce datasets	N/A (review)	Explicitly identifies lack of unified feature-selection + imbalance + interpretability studies

Study	Focus / Venue	Method(s)	Dataset	Metrics	Gap Identified
ECSEL (2026)	Explainable classification via signomial equation learning (arXiv case study)	Signomial equation learning (interpretable model)	UCI Online Shoppers	Accuracy, interpretability metrics	Interpretability studied in isolation, not jointly with imbalance handling

Across these studies, a consistent pattern emerges: papers either optimise for predictive metrics (accuracy, F1, AUC) using increasingly complex models (GNNs, attention mechanisms, ensembles), or they optimise for interpretability using simpler models, but very few examine both dimensions together under varying feature-selection and imbalance-handling regimes on the same dataset. The systematic literature review by the Deep Dive SLR (2025) explicitly confirms this as an open direction, strengthening the case for this project's research question.

4. Research Gap

- Most studies apply a single resampling method (typically SMOTE) or a single feature-selection method in isolation, rather than comparing multiple combinations systematically.
- Performance gains are frequently reported using accuracy alone, which is a misleading metric on an 84.5% / 15.5% imbalanced dataset.
- Models that report the strongest predictive performance (GNNs, attention-based architectures) are the least interpretable, while interpretable models (logistic regression, decision trees) are rarely evaluated under the same imbalance-handling conditions for direct comparison.
- No study identified in this survey jointly evaluates how the choice of feature-selection technique and class-imbalance handling technique together affect both predictive performance and model interpretability on this dataset.

5. Problem Statement

Given session-level browsing behaviour on an e-commerce website, the problem is to predict whether the session will result in a purchase, while explicitly studying how feature-selection strategy and class-imbalance handling technique, individually and in combination, affect (a) predictive performance on the minority (purchasing) class, and (b) the interpretability of the resulting model.

Research Question

How do feature selection and class-imbalance handling affect the performance and interpretability of machine learning models for online purchase-intention prediction?

6. Project Objectives

1. To perform exploratory data analysis on the UCI Online Shoppers Purchasing Intention Dataset, characterising the relationship between session-level behavioural features and purchase outcome.

2. To implement and compare at least three feature-selection strategies (e.g., correlation-based filtering, Recursive Feature Elimination, tree-based feature importance) for this dataset.
3. To implement and compare at least two class-imbalance handling techniques (e.g., SMOTE oversampling and class-weighted learning) against the unbalanced baseline.
4. To train and evaluate four classification models – Logistic Regression, Decision Tree, Random Forest, and SVM – across the resulting feature-selection $\times$ imbalance-handling configurations.
5. To evaluate all configurations using precision, recall, F1-score, and ROC-AUC, with particular attention to minority-class (purchase) recall.
6. To assess model interpretability alongside performance, using feature-importance and SHAP-based explanations, and to identify which configuration offers the best performance-interpretability trade-off.

7. Dataset Description

Attribute	Detail
Source	UCI Machine Learning Repository – Online Shoppers Purchasing Intention Dataset (Sakar & Kastro, 2018), DOI: 10.24432/C5F88Q
Number of instances	12,330 shopping sessions, each from a distinct visitor within a one-year period
Number of features	18 total – 10 numerical, 7 categorical, 1 binary target (Revenue)
Target variable	Revenue (Boolean) – whether the session ended in a purchase
Class distribution	10,422 sessions (84.5%) = No Purchase; 1,908 sessions (15.5%) = Purchase – notably imbalanced
Numerical features	Administrative, Administrative_Duration, Informational, Informational_Duration, ProductRelated, ProductRelated_Duration, BounceRates, ExitRates, PageValues, SpecialDay
Categorical features	Month, OperatingSystems, Browser, Region, TrafficType, VisitorType, Weekend
Missing values	None reported
Relevance	Directly matches the problem statement: session-level behavioural features are used to predict purchase intent in real time, enabling interventions such as targeted offers or exit-intent prompts

8. Proposed Methodology / Workflow

Stage 1: Exploratory Data Analysis

Examine class distribution, and relationships between purchase outcome and features such as PageValues, ProductRelated_Duration, BounceRates, ExitRates, VisitorType, Month, and Weekend, using summary statistics and visualisations.

Stage 2: Preprocessing

Encode categorical variables (Month, VisitorType, Weekend, Browser, OperatingSystems, Region, TrafficType), scale numerical features, and split the data into stratified train/test sets to preserve the original class ratio in evaluation.

Stage 3: Feature Selection (experimental variable 1)

Apply correlation-based filtering, Recursive Feature Elimination, and tree-based feature importance ranking independently, producing three reduced feature sets to compare against the full feature set.

Stage 4: Imbalance Handling (experimental variable 2)

Apply SMOTE oversampling and class-weighted loss independently to the training data, compared against the unbalanced baseline, for every feature set from Stage 3.

Stage 5: Model Training and Evaluation

Train Logistic Regression, Decision Tree, Random Forest, and SVM on every (feature-selection $\times$ imbalance-handling) configuration. Evaluate using precision, recall, F1-score, ROC-AUC, and confusion matrices, focusing on minority-class recall.

Stage 6: Interpretability Analysis

Generate SHAP values and feature-importance rankings for the best-performing configurations, and compare interpretability against the corresponding performance metrics to identify the strongest overall trade-off.

9. Expected Outcomes

- A reproducible pipeline that quantifies how feature-selection and imbalance-handling choices affect purchase-intention prediction on the UCI dataset.
- Empirical evidence identifying which model and preprocessing combination best balances minority-class recall against interpretability.
- A ranked feature-importance profile showing which browsing-behaviour signals are most predictive of purchase intent (PageValues is expected to dominate, consistent with prior literature, tested for robustness here).
- A working prototype that accepts session-level inputs and outputs a purchase-intent probability, demonstrable for the project viva.

10. References

1. Sakar, C. O., Polat, S. O., Katircioglu, M., & Kastro, Y. (2019). Real-time prediction of online shoppers' purchasing intention using multilayer perceptron and LSTM recurrent neural networks. *Neural Computing and Applications*, 31(10), 6893–6908.
2. Wang, W., Xiong, W., Wang, J., Tao, L., Li, S., Yi, Y., Zou, X., & Li, C. (2023). A user purchase behavior prediction method based on XGBoost. *Electronics*, 12(9), 2047.

3. Modeling online customer purchase intention behavior applying different feature engineering and classification techniques. (2023). ResearchGate preprint.

4. Zhou, S., & Hudin, N. S. (2024). Advancing e-commerce user purchase prediction: Integration of time-series attention with event-based timestamp encoding and graph neural network-enhanced user profiling. *PLOS ONE*, 19(4), e0299087.

5. Arora, A., Sakshi, & Gupta, U. (2024). Ensemble learning for enhanced prediction of online shoppers' intention on oversampling-based reconstructed data. In *Proceedings of [Springer conference series]*.

6. Machine learning for predicting online shoppers' purchase intentions. (2024). ResearchGate / conference preprint.

7. Alamuri, O. R. S. G., & Bondalapu, C. K. (2025). Predicting e-commerce purchase intention using machine learning. *ASEAN Journal of Scientific and Technological Reports*, 4(1), 42–58.

8. Tokuç, A. A., & Dag, T. (2025). Predicting user purchases from clickstream data: A comparative analysis of clickstream data representations and machine learning models. *IEEE Access*, 13, 45421–45436.

9. Lin, J. (2025). Application of machine learning in predicting consumer behavior and precision marketing. *PLOS One*, 20(5).

10. A deep dive into feature engineering and class imbalance in customer purchase prediction: A systematic literature review. (2025). *Machine Learning* (Springer Nature).

11. ECSEL: Explainable classification via signomial equation learning. (2026). arXiv preprint – Case study on the Online Shoppers Purchasing Intention dataset.

12. Sakar, C. & Kastro, Y. (2018). Online Shoppers Purchasing Intention Dataset [Dataset]. UCI Machine Learning Repository. https://doi.org/10.24432/C5F88Q